Josh Marsico - Synthetic surface generation from ISO metrics

- Overall objective
  - Quantify how much information ISO 25178-2 areal surface texture parameters carry about a measured surface, by testing whether general-purpose foundation models can regenerate a surface from its parameters alone.
  - Introduce parameters progressively by family
    - height → spatial → hybrid → functional/volume → feature
  - The goal is to identify a potential gap in the standards and better capture discrete features.
- AI agents (plus version numbers) you plan to use - denote your accessibility to them too
  - Claude [Opus 5.5 or Fable 5.1, TBD] - LLM
    - Anthropic API. Pay-per-token; batch API at 50% discount.
    - Primary closed-model.
    - Have a claude.ai Pro subscription, but API billing is separate.
  - GPT [GPT-6] - LLM
    - OpenAI API if budgeted
    - Student ChatGPT subscription.
  - Gemini [3.1 Pro or Argon] LMM
    - Google AI Studio free tier (rate-limited) or paid API
    - Student free subscription available.
  - Open-weight model [TBD by HPC allocation: GLM-5.3-Flash / Qwen3.8-27B / gpt-oss-120b] LLM / LMM
    - run locally on lab HPC with vLLM.
- Input data types and structures
  - ISO 25178-2 areal parameters by family - staggered
    - Height: *Sa, Sq, Ssk, Sku, Sp, Sv, Sz and* Functional/volume: *Sk, Spk, Svk, Smr1, Smr2, Vmp, Vmc, Vvc, Vvv*
    - Spatial: *Sal, Str, Std* and Hybrid: *Sdq, Sdr* and Feature: *Spd, Spc, S10z, Sda, Sha, Sdv, Shv*
    - Spatial sampling and grid size
- Ground truth data
  - The measured heightmap each generation is conditioned on, and its full parameter vector (all families, including those withheld at the current level).
  - Maybe also the PSD and feature count?
  - Running set subset per dataset or not: budget decision
- Draft Prompt script and how each step will be evaluated

  System prompt (fixed across all models and levels):

  You are generating a synthetic areal surface height map. You will be given ISO 25178-2 surface texture parameters measured from a real surface, along with the measurement grid. Produce a surface whose parameters, when measured the same way, match the given values as closely as possible.

Output format, strictly:
Line 1: SCALE <z_min_um> <z_max_um> giving the physical heights that integer 0 and integer 1000 map to.
Lines 2–129: 128 rows of 128 comma-separated integers from 0 to 1000.
No other text, code, explanation, markdown.

User prompt template (varies by family level):

Grid: 128 × 128 points, sampling interval [dx] µm in x and y. The surface was levelled and filtered with [S-filter, F-operation, L-filter] before parameters were computed.

Parameters (ISO 25178-2):
[for each parameter in families 1..n: Name = value unit — one-line definition]

Generate the height map.

Evaluation:

1. Parse validity: correct dimensions, integers in range. Malformed outputs are logged by failure type and excluded from error calculation.
2. The parsed grid is converted to µm, assigned the spatial sampling from the prompt, and passed through the same level and filtering pipeline as the reference surface. The full parameter vector is recomputed.
3. Relative error per parameter, aggregated by family, reported separately for parameters that were provided at that level and parameters that were withheld. The withheld-family error measures what the given families imply about the rest.
4. Structural comparison: PSD, presence or absence of discrete features from ISO 25178-2 segmentation.
5. Each prompt is sent to the model K times (K = 3–5) as independent requests, producing K different surfaces for the same target. Every surface is measured on its own, giving K error values per parameter. Then report the average of those K errors as the result for that surface and level, and the standard deviation as the uncertainty. For every request the model version, temperature, reasoning setting, and token counts are recorded so any run can be reproduced.